

Bayune Lutheran Secondary School

Mathematics Department

Assignment No: 1

SCORE: ___/20 mrks

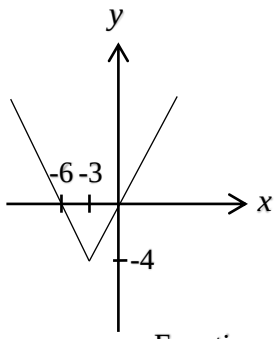
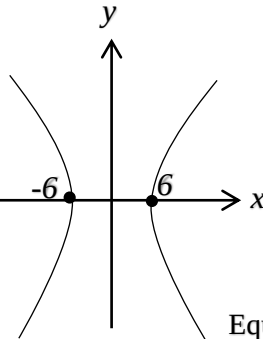
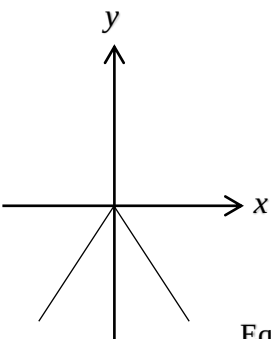
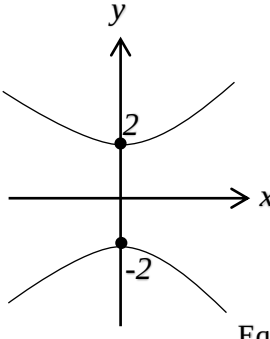
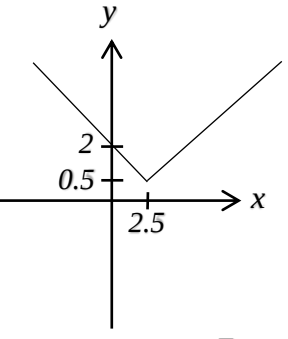
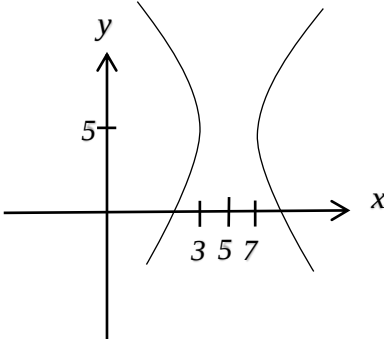
Term 2, week 10 – 2026

Functions & Graphs

Topic: Hyperbolic & Absolute graphs

Surname: _____ Given name: _____ Gender: M/F Class: ___ Date: ___/06/2026

QUESTION 1: Determine the equations of the following graphs:

i) 2 marks  Equation: _____	1 mark  Equation: _____
1 mark  Equation: _____	1 mark  Equation: _____
2 marks  Equation: _____	2 marks  Equation: _____

QUESTION 2: What are the equations of the asymptotes of the hyperbola $\frac{(y-2)^2}{4} - \frac{(x-1)^2}{9} = 1$

Equation of negative asymptote: _____ (1 mark)

Equation of positive asymptote: _____ (1 mark)

QUESTION 3: What is the center coordinate of the hyperbola given by the equation: $\frac{x^2}{25} - \frac{y^2}{144} = 1$?

Center coordinate: _____ (1 mark)

QUESTION 4: Solve the two x values of the absolute equation: $|2x - 3| = 7$:

$x = \underline{\hspace{1cm}}$ or $x = \underline{\hspace{1cm}}$ (2 marks)

QUESTION: Draw the graphs of the equations given below: (2 marks each)

i) $\frac{x^2}{16} - \frac{y^2}{9} = 1$	ii) $\frac{x^2}{18} - \frac{y^2}{50} = 2$	iii) $ x + 4 \leq 9$
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Sighted & Approved by Mx HOD: Mr Gina

Date:16/06/2026